Sagar Kumar Jha

Entry-Level Software Engineer | Systems & Developer Tools
devsagarkumarjha@gmail.com | 8920835343 | Delhi, India
github.com/sagarkrjha | linkedin.com/in/devsagarkumarjha
leetcode.com/u/devsagarkrjha | sagarkrjha.vercel.app

Technical Skills

Languages: C++ (C++20), C, TypeScript, JavaScript; familiar with Rust, Go, Python, Java, SQL
Systems & Storage: LSM-trees, write-ahead logs, SSTables, Bloom filters, content-addressable storage, delta compression, zlib/DEFLATE, SHA-256, CRC-32, POSIX/Win32 APIs
Algorithms & Data Structures: Myers diff, 3-way merge, lowest common ancestor, skip lists, hash indexing, graph traversal
Networking: Git Smart HTTP/HTTPS, pkt-line framing, pack negotiation, libcurl, OpenSSL
Frontend & Tools: React 19, Next.js, Node.js, Electron, Vite, Tailwind CSS, VS Code Extension API, File System Access API, IndexedDB
Build & DevOps: CMake, Docker, GitHub Actions (CI/CD), pnpm workspaces, MSVC, GCC, Clang, Git, GitHub
Databases: PostgreSQL, MongoDB (familiar)

Experience

Independent Software Engineer (Self-Directed, Open Source) July 2024 - Present

- Designed, built and maintained public systems and developer-tool projects end to end (a version control system, an LSM-tree storage engine and a cross-platform knowledge system), covering architecture, implementation, testing, documentation and releases.
- Shipped MiniGit v1.12.0 and CodeShelf v0.3.6 with automated GitHub Actions releases of native Windows, Linux and macOS builds (MiniGit also as a Docker image).
- Wrote 18+ test suites for MiniGit and CI pipelines that build and verify each release across operating systems and compilers.

Projects

MiniGit – Git-Compatible Version Control System | github.com/sagarkrjha/minigit

- Built a C++20 version-control system implementing SHA-256 content-addressed storage, commit DAGs, staging, Myers diff, and LCA-based three-way merging.
- Implemented Git packfile v2/index v2 with delta compression and Smart HTTP v1 transport for clone, fetch, push, and pull; delta-compressed 833 of 1,000 objects in 0.94 seconds.
- Implemented Git Smart HTTP v1 transport for clone, fetch, push, and pull; developed SDKs for Python, JavaScript, Go, Java, and Rust.
- Hardened repository operations with path-traversal protection, repository-boundary validation, and atomic staging-file writes

LSM-Tree Storage Engine – LevelDB-Inspired | github.com/sagarkrjha/level-db-cpp

- Implemented a 32 KB chunked, CRC-checked WAL and skip-list MemTable with arena allocation and acquire/release synchronization.
- Developed SSTables with prefix compression, restart-point indexing, and Bloom filters (10 bits/key) to reduce unnecessary disk reads.
- Implemented MANIFEST-based version tracking, L0–L6 leveled compaction, and tombstone-safe deletion; exposed Put, Get, Delete, WriteBatch, and iterator APIs.
- Developed the read path across active and immutable MemTables and SSTables using Bloom filters, block indexes, and binary search.

CodeShelf – Local-First Developer Snippet Manager | github.com/sagarkrjha/codeshelf

- Built a pnpm monorepo powering web, Electron desktop, and VS Code clients through a shared TypeScript core..
- Implemented offline-first persistence with atomic file writes, browser file-system access, and bidirectional desktop/IDE synchronization; integrated optional Gemini-powered metadata autofill.
- Developed weighted lexical search, VS Code snippet capture and insertion workflows, and Myers-based revision history with rollback; benchmarked 85.9K diff operations/s and 0.91 ms to rank 50 snippets.

Education

Master of Computer Application (MCA), Computer Science July 2026 - Present
Uttaranchal University, India
Bachelor of Computer Application (BCA), Computer Science July 2019 - July 2024
Indira Gandhi National Open University, India

Achievements

- MiniGit selected as a verified reference implementation in the XERJ Corpus Hub for AI coding agents.
- 16+ GitHub stars across open-source systems projects; GitHub Pull Shark badge (x2).